Jesus Rey Gonzalez

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Education

University of Central Florida Orlando, FL

Bachelor of Science in Computer Science *Expected July 2028*

Experience

Software Engineer Intern February 2026 – Present

Florida's Turnpike Enterprise (FDOT) *Orlando, FL*

- Develop customer-facing frontend features for the SunPass tolling platform, public infrastructure serving over **3 million daily** Florida Turnpike users, in **React**, **TypeScript**, and **Tailwind CSS** with **MobX** state management.
- Build cross-platform mobile features in **React Native**, sharing component and state-management patterns across web, **iOS**, and **Android** to reduce duplication and keep behavior consistent.
- Architected, tested, and documented **RPC** backend integrations using **Hoppscotch**, collaborating with cross-functional engineering and business teams to validate design against operational requirements.
- Leverage **AI-assisted development tools** to accelerate root-cause diagnosis of complex frontend defects and scaffold components and unit tests, validating every suggestion against the production codebase.

Projects

Volante | *Rust, Embassy, ESP32-S3, BLE, Python, PyTorch, Swift, wgpu* October 2026

- Built a racquet sensor pod for badminton analytics with async **Rust** firmware on an **ESP32-S3**, streaming IMU data at **500 Hz** over a custom **Bluetooth LE** protocol shared by Rust, **Swift**, and **TypeScript** clients.
- Developed a labeled-data collection pipeline and **stroke classifier** for **10 badminton strokes** from IMU features, with **Python/Swift parity tests** keeping features and predictions identical on iPhone and Apple Watch.
- Created an arena where users train **PPO reinforcement-learning** agents and climb an **Elo** ladder, hosted on a **Raspberry Pi 5** with matches rendered in 3D by a **wgpu** engine compiled to **WebAssembly**.

DronePath | *Python, Google OR-Tools, NumPy, Shapely, Plotly* October 2025

- Placed **3rd out of 188 teams** in the **NextEra Energy** Drone Optimization Challenge at Knight Hacks 2025 with a route-planning tool that minimizes drone flight distance for **power line inspection** across Florida.
- Implemented a multi-drone routing pipeline in **Python** using **Google OR-Tools** with **Dijkstra** and **TSP** graph algorithms under real constraints: battery capacity, airspace boundaries via **Shapely** geofencing, and full power-pole coverage.
- Built an interactive **Plotly** visualization of optimized routes, total distance, estimated energy consumption, and mission count, processing raw **NumPy** geospatial datasets into navigable graph structures.

TermText | *Go, PostgreSQL, WebSockets, Docker, Fly.io, GitHub Actions* June 2026

- Architected a real-time chat application in **Go** with a concurrent **WebSocket** hub supporting rooms, DMs, presence, typing indicators, and read receipts, paired with a **Bubble Tea** terminal UI client.
- Implemented **end-to-end encrypted** DMs using Curve25519/NaCl with **argon2id** password hashing and **PASETO v4** session tokens, plus keyset-paginated history and **PostgreSQL** full-text search via **sqlc**.
- Hardened for production with **Prometheus** metrics, per-IP rate limiting, and structured logging; deployed a 23MB distroless **Docker** image to **Fly.io** with **GitHub Actions** CI.

Leadership & Involvement

Software Engineer February 2026 – Present

Knight Hacks @ University of Central Florida *Orlando, FL*

- Contribute to Forge, an open-source **TypeScript monorepo** powering Knight Hacks' internal tooling, building full-stack features across **Next.js**, **tRPC**, and **Drizzle ORM** with **PostgreSQL**.
- Developed type-safe **tRPC** API endpoints with role-based access control, **Zod** input validation, and Discord webhook audit logging following established authentication patterns.
- Built data visualization components with **Recharts** for the admin analytics dashboard, implementing weekly time-bucketed bar charts to track dues payment trends for club officers.

Technical Skills

Languages: Python, Go, Rust, TypeScript, JavaScript, C, C++, C#, Swift, Java, SQL

Frameworks/Libraries: React, React Native, Next.js, Tailwind CSS, MobX, tRPC, FastAPI, Drizzle ORM, Zod, PyTorch, wgpu, Embassy, NumPy, Google OR-Tools

Databases & DevOps: PostgreSQL, Redis, MongoDB, REST APIs, WebSockets, Docker, GitHub Actions, CI/CD, Git, Linux